Routing, Not Reasoning: A Fail-Closed LLM Agent for Natural-Language Netlist Transformation

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Abstract

Natural-language interfaces to EDA tools must interpret free-form requests without compromising circuit correctness, often with models that cannot be adapted. We present a fail-closed agent for natural-language netlist manipulation that confines the LLM to translating each request into one intent from a closed catalog of predefined operations. A deterministic engine executes all circuit operations, and a mutation is committed only after a proof of functional equivalence and checks of the operation's structural constraints; otherwise it is rolled back. We state the assumptions under which committed netlists remain equivalent and separate this functional guarantee from request conformance, which it does not provide. On a 1,420-request paraphrase stress set for the ICCAD 2026 CAD Contest Problem A, the full intent catalog reaches 88.4% routing accuracy with gpt-4o-mini, against 61.4% for a bare intent list and at most 73.3% for supervised classifiers; claude-haiku-4.5 reaches 97.4%. Fault injection exposed one guard-bypassing operation; after its repair, none of 24 injected faults produced an unsafe commit, while 7 produced equivalent but non-conforming edits.

1. Introduction

Natural-language (NL) interfaces are entering EDA flows, from design question answering to tool-script generation and RTL repair [1]–[3]. Applied to gate-level netlist manipulation, such an interface must meet two requirements at once. First, correctness is not a quality score: a netlist whose function has been silently altered is unusable downstream, which is why production flows gate changes behind equivalence checking. Second, the design state persists: requests arrive in sequence and operate on the netlist as earlier requests left it, so an early error can shape every answer that follows. These requirements often come with limited model capability. Organizations that treat designs as confidential IP may prefer small, locally deployable models [5]; benchmarks such as the ICCAD 2026 CAD Contest Problem A [6] mandate frozen mini-class models; and few teams maintain a fine-tuning pipeline.

When the LLM itself generates the executable artifact — a tool script, an edited netlist, or RTL code — its errors have a direct path into the design. Post-hoc checks catch many of them, but the failure surface is hard to bound, and a smaller model raises the error rate the checks must absorb.

We take the opposite approach: the LLM routes; it does not rewrite. Its only output is one intent object per request, drawn from a closed catalog of about 80 predefined operations with typed parameters. A deterministic engine performs every circuit operation. Parameters that name nets or gates are grounded against the current netlist before execution, and each mutation runs on a snapshot that is committed only after a proof of functional equivalence, obtained through a register-cut projection, and checks of the operation's structural constraints; otherwise the snapshot is restored.

This architecture bounds what the LLM can damage, but not what it can misunderstand. Under explicitly stated assumptions, every committed netlist is functionally equivalent to its predecessor; a mis-routed request can still commit a valid but unwanted edit, and that edit persists into later requests. We therefore evaluate both sides: how accurately small frozen models route requests, and which faults the guards prevent and which they cannot.

Our contributions are:

- **Architecture.** A constrained interface for NL netlist manipulation that separates LLM intent interpretation from deterministic execution and proof-gated commit, with an explicit statement of what it guarantees (functional integrity, under stated assumptions) and what it does not (request conformance) (§3).
- **Routing evaluation.** A controlled study on 1,420 paraphrased requests that compares a bare intent list, a full catalog, retrieval configurations, model choices, and supervised baselines, and analyzes the near-synonym operations behind most residual errors (§4.2–4.3).
- **Validation and limits.** Fault injection and the official contest evaluation, quantifying prevented violations, residual semantic drift, and one guard-bypassing operation that the experiment exposed and that we repaired (§4.4–4.5).

2. Problem Setting and Related Work

Setting. A session loads one flat gate-level Verilog netlist — two-input primitive gates, inverters, buffers, and D flip-flops with asynchronous control — and then receives free-form NL requests, one per line, answered in lock-step [6]. Requests analyze (depth, cones, paths, equivalence), edit (buffer insertion, basis conversion, renaming), or optimize (depth, gate count) the current netlist. Hard requirements — functional equivalence where requested, stated structural bounds, gate-basis restrictions — are all-or-nothing: violating any one zeroes the testcase.

LLM agents for EDA. Script-generating agents emit tool scripts for EDA flows and check them before execution [1], [7], [8]; flow-tuning agents search a bounded parameter space [9], [10]; FluxEDA keeps backend instances alive so that later steps observe earlier state [11]. RTL-oriented work generates or repairs code against compilers and testbenches [2]–[4], and recent benchmarks target netlist manipulation and tool calling [12], [13]. These systems differ in how much execution freedom the LLM holds; in ours it holds none beyond selecting one predefined operation and its parameters on a persistent gate-level netlist.

Structured tool calling. Schema-constrained function calling is well established [14]–[16], typically retrieving tool documentation into the prompt. We instead retrieve labeled example requests, keep the full catalog in every prompt, and validate name-typed parameters against a design state that earlier requests have modified.

Verified and checked execution. LLM-Modulo argues for LLM generation behind external verifiers [17]; obligation and transaction checks constrain agent actions outside EDA [18]–[20]. Concurrent work on the same benchmark shows that a run passing every equivalence check can still contain incorrect answers and violated constraints, and adds request-derived completion checks [21]. That work checks conformance, whereas ours bounds functional damage; the two are complementary. Our contribution is therefore not a new verification primitive but a commit protocol for persistent sequential netlists, state-aware parameter grounding, and a measured account of what the combination prevents.

3. Fail-Closed Agent Design

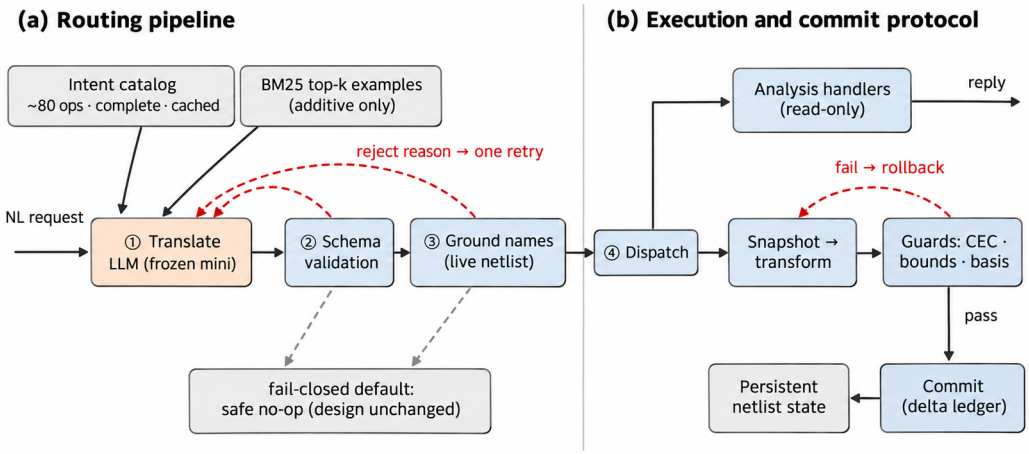


Fig. 1. The fail-closed agent. (a) Routing (§3.2): the LLM only translates; schema validation and name grounding against the live netlist return a rejection reason for one corrective retry (dashed red), and every failure exit ends in a safe no-op (dashed gray). (b) Commit protocol (§3.3): mutations run on a snapshot that is committed only if all checks pass; otherwise it is rolled back.

3.1 Closed-world intent interface

The catalog defines about 80 intents covering analysis (counts, depth, cones, paths, connectivity, functional properties), transformation (basis remapping, buffer insertion, cleanup, renaming), and optimization (depth, gate count). Each intent declares required and optional parameters with types; enumerated parameters such as the gate basis have closed value sets. The catalog text — a description of each intent plus disambiguation entries for operations that users phrase similarly — is part of every prompt as a fixed, cacheable prefix of about 12.8k tokens. Retrieval only adds examples and never removes catalog entries, so a retrieval miss cannot put the correct operation out of the model's reach.

The operation space is closed; the parameter space is not. Which net, cone, bound, or basis an operation applies to is decided per request. A request that maps to no catalog operation yields a safe no-op that reports the design unchanged. Extending the catalog means adding a handler, a schema, and a disambiguation entry — an auditable engineering change rather than model retraining.

3.2 Routing with live-state grounding

Routing proceeds in four stages (Fig. 1a). (1) *Translation:* the LLM receives the catalog and the 50 most similar labeled requests, retrieved by BM25 from a bank of 940 requests drawn from the contest's public and beta rounds, and emits one JSON intent object. (2) *Schema validation:* an unknown intent, a missing parameter, or an illegal enumerated value is rejected, and the rejection reason is returned to the LLM for one corrective retry. (3) *Name grounding:* every parameter that must name a net, gate, or flip-flop is checked against the current netlist, which earlier requests may have modified; a nonexistent name triggers re-translation with the error message. (4) *Dispatch* to a deterministic handler: analysis handlers run graph algorithms on the netlist IR, and mutation handlers enter the commit protocol of §3.3. Translations are cached within a session, and if every stage fails the agent returns the safe no-op.

3.3 Guarded execution and commit protocol

A mutation handler transforms a snapshot of the netlist (Fig. 1b), and three checks decide whether the snapshot replaces the current design.

1. **Functional equivalence.** A register-cut projection turns the sequential netlist into a combinational one: each flip-flop's D, CK, RN, and SN pins become pseudo-outputs and its Q pin a pseudo-input. Flip-flops correspond by instance identity, and a change to the register set is rejected outright. Because every flip-flop input is observed, combinational equivalence of the projections implies sequential equivalence for the testbed's flip-flop model, including logic on clock and asynchronous pins. ABC's `cec` decides equivalence [22], with a yosys check as a fallback.
2. **Structural bounds.** Bounds carried by the operation, such as a maximum fanout, are recomputed on the result.
3. **Basis purity.** If the operation restricts gate types, the gate-type set of the result is checked.

The acceptance rule is fail-closed: the snapshot is committed only when every check passes and equivalence is proved. A disproof, a timeout, or an inconclusive result — for instance on a malformed netlist — restores the snapshot. Candidates produced by the optimizer, including precomputed candidates that serve as search seeds, obey the same rule. Committed mutations are recorded in a delta ledger that later requests ("how many buffers were added?") query.

3.4 Scope of the guarantee

We distinguish three properties of a committed edit: *functional integrity* (the netlist computes the same function), *structural validity* (it meets the bounds and basis of the routed operation), and *request conformance* (the routed operation is what the user asked for). The architecture targets the first two, under three assumptions:

- **(A1) Complete mediation.** Every operation that mutates the netlist, including renaming, executes through the commit protocol.
- **(A2) Sound checking.** A proof of equivalence reported by the checker is correct for the projected circuits.
- **(A3) Faithful projection.** The register-cut model matches the flip-flop semantics of the netlist.

Under A1–A3, every committed netlist is functionally equivalent to its predecessor and structurally valid for the routed operation. Request conformance is not guaranteed: a mis-routed request can commit an equivalent but unwanted edit, and because the design persists, that edit changes what later requests observe — for example, a gate count or a net name that a later question depends on. The guards also enforce only the requirements carried by the current operation, not constraints established by earlier requests. A1 is the assumption most exposed to implementation error, and §4.4 tests it directly.

4. Evaluation

4.1 Setup

Requests. We evaluate routing on a stress set of 1,420 requests that our team wrote during the contest by paraphrasing the official request patterns: 71 operations with 20 paraphrases each, of which 1,065 use deliberately indirect or unusual wording and 355 use standard phrasing. The set provides ground-truth intents and parameters but no netlists, so we score the translation output directly. The retrieval bank of 940 requests from the contest's public and beta rounds is disjoint from the stress set. A second, high-recall bank of 2,130 author-written paraphrases contains paraphrases of the same operations as each test request; we use it with leave-one-testcase-out (LOCO) exclusion and only as an upper bound, because it shares the stress set's distribution. Since the catalog's disambiguation entries were refined during the contest while inspecting errors on this set, accuracies on it may be optimistic.

Metric and models. The primary metric is intent accuracy: whether the selected operation is correct. We use the two contest-mandated models, gpt-4o-mini and claude-haiku-4.5, at temperature 0.2, and repeat the gpt-4o-mini headline settings five times (N=5). Evaluation rounds run strictly one at a time, and a round with any degraded API call is discarded: we once observed rate limiting silently convert LLM calls into no-op responses and depress a measured accuracy from 97.8% to 67.2%.

4.2 Routing accuracy

Table 1. Intent accuracy on the 1,420-request stress set. Ranges span N runs. †Upper bound: the bank shares the stress set's distribution.

Router	Labeled data used	Intent acc.	N
BM25 nearest neighbor	production bank	47.4%	—
SBERT embedding + kNN	all labeled requests (LOCO)	72.8%	—
TF-IDF + linear SVM	all labeled requests (LOCO)	73.3%	—
gpt-4o-mini, bare intent list	—	61.4%	1
claude-haiku-4.5, bare intent list	—	78.7%	1
gpt-4o-mini, full catalog	—	88.4% (88.3–88.7)	5
gpt-4o-mini, catalog + retrieval (k=50)	production bank	88.9% (88.6–89.2)	5
gpt-4o-mini, catalog + retrieval (k=50)†	high-recall bank (LOCO)	95.1 / 95.6%	2
claude-haiku-4.5, catalog + retrieval (k=50)	production bank	97.4%	1

Catalog information is the largest lever. A bare list of intent names yields 61.4% with gpt-4o-mini; the full catalog — descriptions, parameter schemas, and disambiguation entries together — raises it to 88.4% (+27.0 points). The bare condition keeps the schema-feedback retry, whose fixed message mentions class-level depth intents, and contains no other catalog text. The gap between the two models is 17.3 points with the bare list and 8.5 points with the full pipeline, so the catalog helps the weaker model more.

Supervised baselines underperform. With every labeled request available, the best classifier reaches 73.3% (TF-IDF features with a linear SVM), and frozen sentence embeddings reach 72.8%. Classifiers and LLMs are scored on the same operation-level metric; the LLM additionally fills in parameters (84.6% parameter accuracy for gpt-4o-mini with the catalog, 94.1% for claude-haiku-4.5 with retrieval), which the classifiers do not attempt. With only the production bank, three operations have no examples and are unreachable for nearest-neighbor routing, whereas the LLM reaches them through the catalog.

Retrieval value depends on the bank and the model. With the production bank, retrieval adds only 0.5 points for gpt-4o-mini, and retrieval depth barely matters: k from 5 to 100 spans 88.1–89.4% without a monotone trend, while label recall rises from 60% to 86%. The high-recall bank adds 6.4 points (an upper bound), and claude-haiku-4.5 reaches 97.4% on the production bank, whose label recall at k=50 is only 82.1%. A stronger model thus needs less help from retrieval; for the weaker model, bank quality matters, and its effect (+6.4 points) is comparable to switching models (+8.5 points at a fixed configuration).

4.3 Failure analysis

Moving gpt-4o-mini from the production to the high-recall bank removes 84% of give-ups (44 to 7) but only 43% of wrong-operation picks (110 to 63): good examples mainly help the model recognize unfamiliar phrasing. What remains is dominated by confusions between semantically similar operations, such as primary-input-to-output depth versus global maximum depth, or floating versus dangling nets, and part of it reflects genuine ambiguity in the requests. The supervised baselines confuse the same operation families. The concurrent agent of [21], built on a different architecture, reports the same kind of scope error on this benchmark (a pairwise depth of 10 reported as the whole-design depth of 53). On this request distribution, near-synonym discrimination is a major source of residual routing errors.

4.4 Fault injection

To test A1 and the guards, we corrupt routing output at the dispatch layer, simulating worst-case LLM behavior: numeric bounds off by one in either direction, a wrong but implemented basis or transformation, a basis outside the closed world, a target redirected to a different existing net, a swap to a different intent, and a function-changing rewire injected inside the engine. We run 24 injections over six transformation-heavy public testcases (the six uncorrupted runs pass all checks) and classify each final netlist against the hard requirements of the original request (Table 2).

Table 2. Fault-injection outcomes (24 injections; classes are mutually exclusive).

Outcome	Meaning	Before fix	After fix
Rejected	refused before execution (closed-world value or name collision)	2	5
Rolled back	equivalence disproved; snapshot restored	3	3
Benign	corrupted edit still met every hard requirement of the request	9	9
Drift	equivalent and valid for the routed operation, but non-conforming	7	7
Unsafe	non-equivalent netlist committed	3	0

Only rejections and rollbacks are detections; the nine benign cases passed because the corrupted edit happened to remain acceptable. The three unsafe commits shared one cause, a violation of A1: renaming was treated as cosmetic and bypassed the commit protocol, so renaming a net onto an existing name merged two drivers while the agent reported success. In one of these cases ABC returned an inconclusive result on the double-driven net, which an evaluation harness that treats inconclusive results as passing would have accepted. After we added a check that rejects renames onto existing names, no injection produced an unsafe commit, and the 40 public testcases pass all 84 of their hard-requirement checks. The seven drift cases mark the conformance boundary of §3.4: the guards enforced the corrupted operation's contract rather than the user's sentence, and the response claimed success. Detecting them requires request-level completion checks such as those of [21].

4.5 Official evaluation and post-contest analysis

In the contest's final round (100 testcases and 674 requests, unseen during development), the submitted configuration met every hard requirement and scored 671.366 of 674 under official scoring. All losses came from optimization requests, which are scored by their cost relative to the best cost among all entries. Table 3 compares the official result with two post-contest re-evaluations of the same 46 optimization requests.

Table 3. Optimization score on the 46 final-round optimization requests.

Run	Score (/46)	Compute
Official evaluation	43.366	1 run, 300 s per request
Re-run, same configuration	43.514	1 run, 300 s per request
Best of up to four runs	43.983	up to 4 runs, including an earlier engine version

A single re-run under the same per-request budget improves 15 requests and worsens 4, for a net gain of 0.148: the optimizer's search is budget-bound and stochastic, and one run is far from its best achievable result. Taking the best of up to four runs, at up to four times the compute, adds 0.617; all 17 improved netlists were re-verified independently with ABC, and four of them beat the best cost among all contest entries. These figures are post-contest upper bounds, not a comparison under the official protocol. Three requests resist re-sampling entirely (1.41 points) and call for algorithmic improvement.

5. Limitations and Conclusion

Limitations. We evaluate on one benchmark. The routing stress set was written by the authors and the catalog refined against it, so its accuracies may be optimistic; the official final round provides an out-of-sample, end-to-end check. The cloud mini-class models stand in for the small, locally deployable models that motivate this work, and direct validation on open-weight models remains future work. Some settings were run once (N=1). The functional guarantee rests on A1–A3 and on a correct implementation of them, and drift — valid but non-conforming edits — is outside its scope.

Conclusion. Restricting the LLM to selecting predefined operations, and committing their results only on proof, bounds the functional damage that an LLM error can cause in persistent netlist manipulation, under assumptions that can be stated and tested; fault injection found and closed a gap in the assumption most exposed to implementation error. What the architecture does not bound — whether the executed operation is the one the user meant — remains, and our routing study locates most of it in near-synonym operations. For agents that edit an existing design, safe execution and request conformance are separate problems, and each needs explicit treatment.

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